



MINING DEALER BEST PRACTICE

Leveraging PLESN Data at the Dealership

**Maintenance and
Repair**

Application

**Component Life
Management**

**Component
Renewal (CRC)**

**MARC
Management**

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1.0 Introduction

Demand planning for key replacement parts & components is critically important as the Cat dealer works with the local mine site to achieve machine availability targets as well as other key mining metrics. In order to properly plan for upcoming parts needs at the site, many dealers have been using elements of the Planned Component Replacement (PCR) process, a process in place since approximately 2005, to understand what, when, and how many major components will be required for the Traditional Mining machines operating on sites in their territory. And given the parts commonality typically inherent with traditional Cat machines, the PCR process has worked well in preparing the enterprise supply chain (both the dealer and Caterpillar) to ensure the right mining parts are available at the right time and delivered to the right location to enable machine uptime.

However, with the introduction of Expanded Mining machines into the Cat mining product portfolio, it was observed the PCR process may not be sufficient in providing an accurate depiction of what key replacement parts & components are required. One of the primary drivers for this is due to the lack of parts commonality that typically exists in the Expanded Mining machine models, as these machines can be more custom in nature and have parts lists that are not widely shared even within the same product line. Therefore, in order to ensure the correct parts are being planned for by the site, dealer, and Cat, the Plan for Every Serial Number (PLESN) initiative was launched at the beginning of 2015. Not only does PLESN provide greater insight into the correct key replacement parts & components that will be needed for an upcoming rebuild or replacement event, but PLESN also helps drive:

- Increased communication between the site, dealer, and Cat on the current & desired repair strategy for a specific mining machine
- Accurate identification of future parts & service sales opportunities
- Better understanding of the required parts, labor, and tooling that will be required to service the machine and maximize uptime
- Improved mining parts and machine availability to achieve customer satisfaction & expectations
- Enablement of the Seed / Grow / Harvest model to achieve POPS-C and OPACC targets.

2.0 Best Practice Description

This Best Practice describes the internal processes that Finning Canada is implementing to better leverage and take action on PLESN data. The primary areas of focus for this Best Practice are: roles & responsibilities; process used by Finning's mining parts supply chain team; and the process used by Finning's mining parts commercial team.

3.0 Implementation Steps

Before beginning on a successful journey with the Plan for Every Serial Number, it is imperative the dealer's senior leadership team are committed and engaged, and are supportive of the effort at the outset. The PLESN initiative crosses functional areas at the dealership, is resource-intensive, and does require much hard work in order to develop and stabilize the necessary processes. Without this level of engagement and commitment, it is not likely to be successful. Similarly, Cat region leadership (both in the Global Aftermarket Solutions Division [GASD] and Cat Global Mining [CGM]) must also fully support and proactively engage in the PLESN initiative

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to ensure its success. Through the collective commitment from the dealer and Cat, PLESN can withstand the challenges that will be faced as the hard work begins.

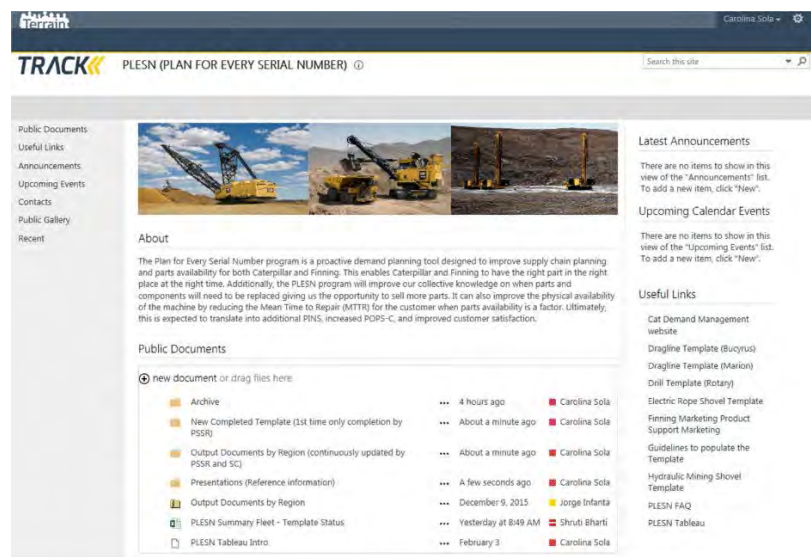
After commitment to the initiative has been given, a process map for how PLESN will functionally and tactically work should be developed. Cat has a high-level demand planning process map that can be leveraged, but the dealer should then customize the details within that high-level map to ensure local considerations are taken into account. Please see the attached file 'PLESN Process-V2-Final-April 8 2016.pdf'. The file represents the current process in place at Finning Canada to action the data. To reiterate, however, each dealer should localize the process map to incorporate any relevant local nuances.

Once the process map has been established, a Roles & Responsibilities (R&R) document can be created to ensure key departments, and individuals within those departments, are identified for the process. This helps ensure each understands his or her responsibility in the PLESN process, and how his or her work contributes to the success of the PLESN initiative in order to fully maximize its potential. An example of the R&R can be found in the supporting attachment titled 'PLESN Required Data and R&R (April 2016).pptx' for reference.

As these implementation steps are completed, communication throughout the organization is critical to ensure people are aware of what is happening, what the change means to them, who they can turn to for questions and clarity, and how they can support the endeavor. For example, Finning developed both an internal dealer document, as well as an external customer document, to ensure the initiative was fully communicated and understood by those who would be impacted. By more thoroughly communicating with key individuals and groups, Finning was able to gain better engagement both inside and outside the dealership.

Additionally as part of the internal communication strategy, Finning developed an internal site that can be accessed by dealer employees to not only learn more about the PLESN initiative, but access important documents such as PLESN FAQ's, presentations, templates, status updates, etc. A screen print of this internal site can be found below (see Exhibit A).

Exhibit A:



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Also, Finning leveraged the Tableau software to be able to effectively and efficiently summarize and share completed PLESN data results received from the account representatives. To explain the Tableau screen (see Exhibit B below), Finning customized the report to include the following areas:

- Upper right quadrant: provides a high-level, consolidated view of SKU (Stock Keeping Unit, or line item) and the associated extended dealer net opportunity by the probability of sale %
 - o This gives the dealer a quick view of the total dealer net opportunity from the completed plans submitted by the account representatives, and the associated probability of closing those opportunities as won.
- Upper left and middle quadrants: provides an understanding of what machines are active compared to the total population by region (left quadrant) and how many PLESN's have been completed compared to the total active population by region (middle quadrant)
 - o This provides the dealer a quick glance at how many machines are active in a given region, and also how many of those active machines in the region have a completed PLESN
- Lower left quadrant: is a high-level consolidation of SKU and the associated extended dealer net opportunity by % to PCR category; those within the 0% to 74% to PCR range are considered "On Time"; those within the range of 75% to 99% are considered "Close to Replace"; those over 100% are considered "Overdue"
 - o This quantifies for the dealer the various opportunities coming due and the value of the opportunities that need immediate or near-term attention
- Lower right quadrant: shows how many SKU are coming due for order and when the order should be placed (based on operational hours being forecasted as well as the Cat lead time to ship the part to Finning (from order to arrival)), as well as the current stock levels for Finning & Cat
 - o This provides Finning a quick snapshot of the volume of orders that will need to be placed on Cat to ensure the necessary parts are in the correct location when needed to support the rebuild / replace / repair event

Exhibit B:



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4.0 Benefits

There are multiple benefits, both for the customer and the dealer, that result from a robust Plan for Every Serial Number process. First, concerning the customer, the following benefits should be realized:

- Higher machine availability and therefore productivity due to understanding the what part is needed, when it is needed, and where it is needed
- Thorough understanding by the dealer and Cat of the customer's mining machine, its current condition, and the repair strategy
- Improved parts availability and therefore a reduction in emergency freight and fees due to future demand signals being provided to the dealer & Cat, and associated action being taken when and where necessary
- Unavailability metric due to parts delays for component replacement should be minimized as proper planning & scheduling takes place between the site, dealer, and Cat
- PCR compliance metric should be maximized through the coordination and synchronization between the site, dealer, and Cat of what replace / rebuild / repair events are expected to occur and when they are expected to occur

Similarly, there are also many benefits to the dealer that flow from a solid PLESN process in place. Some of these include:

- Improved visibility and understanding of parts & service opportunities coming due with key mining customers & sites
- Potential to grow parts & labor sales and POPS-C (Percentage of Parts Sales – Caterpillar (equipment)) with mining customer base
 - o As an example, Finning was able to develop an effective proposal / quote and then close a ~\$4 million opportunity for Expanded Mining ERS (Electric Rope Shovel) undercarriage through an opportunity highlighted by the PLESN effort
- Improved parts margin & OPACC (Operating Profit After Capital Charge) by lowering emergency / expediting freight and fees through improved demand planning
- Improved coordination of planning across departments (Parts, Product Support Marketing / Sales, Service, CRC (Component Rebuild Center)) at the dealership regarding future product support business
- Change from reactive position responding to the customer's demands to a proactive position
 - o Dealer is able to engage the customer early to discuss options rather than after failure or after the customer has decided on a competitive alternative
- Opportunity to further the Seed / Grow / Harvest model and strengthen the dealer's PINS (Percentage of Industry Sales) position through customer product support satisfaction
- Better position of PSSR (Parts & Service Sales Representative) / Account Representative as proactive consultant to the customer
 - o By leveraging the Tableau software, Account Representatives can drill down to a specific machine serial number / part number to understand what is coming due and when; this allows the Account Representative to better plan for the next visit to the site, understanding what opportunities need to be quoted and/or closed to allow for the necessary order processing and lead time to have the part ready when required by the customer

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5.0 Resources Required

In order to develop an effective PLESN process at the dealership, resources are required, primarily in the area of personnel. Although Finning has invested in electronic resources / systems to better facilitate the aggregation, summarization, and communication of PLESN, this is not a requirement. However, active engagement from key personnel is critical to PLESN's success.

The roles & responsibilities of the dealership and Cat are found in Exhibit C below. These roles & responsibilities should consistently be part of the PLESN process if the initiative is to be successful and achieve the aforementioned benefits. These roles & responsibilities can also be found on the Mining Demand Management site at:

<https://dealer.cat.com/miningdemandmanagement>.

Exhibit C:

Dealer		Cat	
Function	Primary Activity	Function	Primary Activity
Marketing	- Update machine population status by serial number - Update SMU and utilization by serial number - Qualify & develop parts & service sales leads from PLESN	Aftermarket Solutions Rep	- Support dealer PLESN completion - Analyze merchandising support requests based on lead development, if needed
Mining Operations	- Conduct MSR with site to understand / set M&R strategy for machine - Customize dealer repair option file from Cat centerline (Builder)	GASD Mining Demand Planner	Coordinate PLESN effort / completion in region
PSSR or Account Rep	- Complete template & PLESN for site / machine serial number - Validate PLESN data to ensure accuracy - Refresh PLESN every cycle (quarterly)	Mining Product Support Rep	- Support MSR with dealer & site - Support PLESN completion
Parts Dept.	- Collect & analyze the PLESN data - Analyze current inventory position (on-hand & on-order) - Communicate PLESN results - Place parts order(s) based on analysis / need	Field Sales Support Engineer	- Support template pre-population - Support PLESN completion
Service Dept. and/or CRC	- Analyze labor and tooling planned vs. required based on PLESN data - Support development of dealer repair option file	Mining Parts & Product Support Team	- Administer PLESN program - Aggregate dealer PLESN files - Share dealer PLESN files with internal business partners (CLS; MBD) - Communicate PLESN aggregation results
Dealer Site Manager	- Ensure localized dealer repair option exists for site's mining serial numbers - Ensure validated updates communicated back through dealer & Cat channels - Ensure unavailability metric minimized from parts delays	Cat Logistics Team (CLS)	- Compare on-hand & on-order vs. PLESN signal - Take material requirement action as needed based on analysis
		Mining Machine Business Division (MBD)	- Provide part number interpretation as needed - Take action on PLESN signal as needed for ETO / DTO items (qualify & scope requirements; analyze for upgrade or retrofit potential) - Analyze site-specific life cycle data from PLESN - Analyze probability of sale from PLESN
		Site Performance Manager	- Ensure localized dealer repair option exists for site's mining serial numbers - Ensure validated updates communicated back through dealer & Cat channels - Ensure unavailability metric minimized from parts delays

6.0 Supporting Attachments / References

- PLESN Process-V2-Final-April 8 2016.pdf
- PLESN Required Data and R&R (March 2016).pptx

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7.0 Related Best Practices

Planning & Scheduling –	1007.2.0-1104
Parts Management –	1007-2.0-1105
Component Management –	1007.2.0-1106

8.0 Acknowledgements

Cat Global Mining would like to thank personnel at Finning Canada for the many contributions to this Best Practice.

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- Jorge Infanta (Supply Chain Supervisor)
- Michael Fradgley (Supply Chain Manager)
- Carolina Sola (Expanded Mining Parts Commercial Supervisor)
- Craig Curran-Morton (PLESN Project Manager)
- Rob Hodgson (Director - Expanded Mining Products)

If a Cat dealer would like to discuss Finning's PLESN process further, they are welcome to contact:

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Without Finning's strong engagement and focus on the PLESN initiative, this Best Practice would not have been possible.

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